

BHARAT REFINERIES & PETROCHEMICALS CORPORATION
CENTRAL INSPECTION & NON-DESTRUCTIVE TESTING DIVISION

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ANNUAL COMPREHENSIVE ULTRASONIC INSPECTION REPORT: 2026
Document Ref: BRPC/INSP/AVDU-02/UTG-402 Date: 15-JAN-2026
Refinery Unit: Atmospheric Vacuum Distillation Unit (AVDU-02)
Equipment: Fractionator Column C-101 Overhead Vapor Circuit
Piping Line Identifier: 10-HC-402-CS300 (NPS 10 Carbon Steel)
Design Standard: ASME B31.3 / API 570 Class: 300#
Design Pressure: 258 psig (18.2 kg/cm2g) Design Temp: 320 F

- EXECUTIVE SUMMARY & ULTRASONIC GAUGING FINDINGS:
- 1. Nominal As-Built Wall Thickness: 9.53 mm (0.375 in, Sch 40)
 - 2. Specified Minimum Corrosion Allowance (CA): 2.00 mm
 - 3. Calculated ASME B31.3 Minimum Wall Thickness (t_min): 7.20 mm
 - 4. Retirement Wall Thickness Limit (t_ret): 6.80 mm

ULTRASONIC THICKNESS GAUGING (UTG) RESULTS AT CRITICAL TMLs:

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TML No.	Inspection Location	Thick (mm)	Status
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TML-01	Column C-101 Top Nozzle	8.92 mm	NORMAL
TML-02	Overhead Elbow E-01	8.10 mm	NORMAL
TML-03	Safety Valve PSV-402 In	7.95 mm	MONITOR
TML-04	Elbow E-04 Weld W-12	7.15 mm	DEFICIENT
TML-05	Accumulator V-104 Inlet	8.45 mm	NORMAL
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- CRITICAL DEFECT ANALYSIS (TML-04 / ELBOW E-04):
- Measured Wall Thickness: 7.15 mm (BELOW MINIMUM DESIGN 7.20 mm)
 - Historical Thickness (2024 Inspection): 8.05 mm
 - Metal Loss over 24 Months: 0.90 mm (Corrosion Rate: 0.45 mm/yr)
 - Accelerated localized pitting corrosion observed near weld seam W-12 caused by ammonium chloride salt deposition and wet H2S condensation.

- MANDATORY RECOMMENDATIONS UNDER REFINERY SOP-REF-402:
- 1. Issue Immediate Alert Notice Level 1 for Line 10-HC-402.
 - 2. Prepare formal PSU Approval Note for Chief General Manager (CGM).
 - 3. Perform ASME Sec IX qualified structural weld overlay within 30 days.
 - 4. Recalibrate and verify set pressure on Safety Relief Valve PSV-402.
 - 5. Conduct follow-up ultrasonic testing in 90 days to verify integrity.
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Inspected By: Er. R. K. Sharma (NDT Level III - UT/VT/MT)
Verified By: Er. S. Sengupta (Chief Inspection Engineer)
Classification: STRICTLY CONFIDENTIAL - AIR-GAP ON-PREMISE ONLY

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